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# Introduction

## Research Background

Gait analysis is considered one of the most fundamental topics in biomechanics that has implications in healthcare and diagnosis, physical therapy and sports training, robotics, and human-technology interaction. New bio-movement capturing technologies and machine learning algorithms have boosted the levels of accuracy in the evaluation of gait patterns. GaitPhase is introduced, which contains gait information for 21 subjects, and from which all kinds of classification models can be created. The proposal consists of a

complete kinematic dataset recorded at various walking speeds that will be very useful when analyzing and classifying patients.

## Aim and Objectives

The main focus of this project is to build the machine learning classifiers and later assess the efficiency of the classification of gait data. The specific objectives are:

1. To adopt gait data from GaitPhase database and get these data ready for analysis.

2. To use classifiers for machine learning such as SVM, Naïve Bayes, Decision Tree, Random Forest, Logistic Regression, and KNN.

3. In order to measure the performance of these classifiers using given standard measures including Accuracy, Precision, Recall and F1-Score.

4. The best goal for this project is going to be to determine the best classifiers for distinguishing between various gait speeds and patterns.

## Research Problem Statement

It is not an easy task to classify the gait patterns because human movement is quite variable and the dimensions of the kinematic data are high. Conventional techniques are usually not accurate to a high degree especially when they are trying to differentiate between different walking speeds. This project does this by employing state of the art machine learning methods to train high quality classifiers that can deal with the complexities of gait data. The attention is paid to the comparison of several classifiers in order to understand which approach is better to use for accurate and reliable classification of the patient’s gait.

## Research Rationale

This view is valid due to the rise in the number of diseases affecting gait and the search for methods for their efficient diagnosis. Another strategy that machine learning provides is to automatically classify problems and increase the accuracy of the classification. Through comparing different classifiers of a selected project, this study will determine the best approaches for gait analysis so as to support the creation of efficient diagnosing and rehabilitation means. Not only, it has extended the knowledge of biomechanics but it also has relevance to the various fields including sports science, ergonomics and human computer interaction field.

# Literature Review

## Comprehensive Models for Various Gait Analysis Techniques

Gait analysis has become something more advanced especially with the incorporation of models and techniques in an attempt to enhance the result's accuracy. Prior to the modern emergence of machine learning predominantly, the more classical approaches were based on feature engineering and manual construction of features to be fed to the classification algorithms. These methods are still used because they are easier and computationally less intensive. However, they are not ideal in solving different kinds of problems, particularly in cases where problems are deeper and complex.

Modern development trends have been toward learning-based methods, based on deep learning algorithms. These approaches including CNNs and RNNs are seen to produce better results when it comes to HAR. In the final review by Sargano et al. (2023), the work also discusses the advantage and disadvantage of handcrafted and learning-based action representation and affirmatively stresses the use deep learning methods for better and faster gait analysis.

## Machine Learning Classifiers for Gait Classification

The use of ML classifiers in gait classification has attracted many researchers as seen in the subsequent literature review. Other classifiers such as Support Vector Machines, Naive Bayes, Decision trees, Random forests, Logistic regression, and K Nearest Neighbors have been used to improve the performance of gait classification.

An empirical analysis conducted in the "Biomechanics" journal in 2024 showcased the utilization of these classifiers for inferring various gait conditions. The researchers employed the sophisticated kinematic data to fine tune and test the classifiers where it was seen that the ensemble classifiers such as Random Forest generally perform better than individual classifiers since the former is less prone to overfitting because of its aptitude to handle high dimensionality.

Another research was dedicated to the concern how to design a hybrid forecasting model based on statistical techniques coupled with deep learning structures. Although this is explained well in a paper by researchers in "Applied Sciences" (2024), it shows that hybrid models, or models that can deal with linear and non-linear parts of gait data, enhance the effectiveness of gait classification systems by ameliorating the last methods' frailties or shortcomings.

## Previous Studies and Findings in Gait Analysis

The following are some of the studies that helped address the current knowledge on gait analysis and its uses: For example, the study by Vickery-Howe et al. (2024) entitled

‘Biomechanics’ aimed at the comparing the physiological, perceptual and biomechanical aspects of male and female gait under different conditions. This work brought useful information about gender dimension in gait analysis and the need to target male clients in gait intervention.

Furthermore, systematic review studies on HAR have highlighted the shift of focus from conventional feature engineering to deep learning approaches. These reviews have made clear key datasets, issues, and opportunities for research in human gait, particularly, they pointed out the complexities of human gait to be addressed by models.

In conclusion, it can be said that the addition of advanced machine learning classifiers and the integration of combination models in gait analysis enabled enhanced and more precise classification systems. Such advancements are bringing hope in the field of diagnosis, both in clinical and rehabilitation, and in sports science where proper gait assessment is important.

# Methodology

## Data Collection Method

The data for this project will be collected from the Kaggle that encompasses the comprehensive data of 21 subjects. This data base encompasses kinematic data recorded with a Qualisys motion capture system and an instrumented Bertec split-belt treadmill. The motion capture system consists of 8 Oqus cameras acquiring at 200 Hz for accurate three dimensional kinematic measurements. The treadmill incorporated force plates where the sampling rate was set to 1000Hz used in measurement of GRCF. These systems are integrated to work in parallel to another to minimize on data discrepancy and less accurate results.

Every subject in the study walked at twelve different speeds and the speeds were as follows, 0. 6 m/s to 1. 7 m/s with step 0. 1 m/s. Data collection was done by capturing the respondents walking at three speeds for one minute each and amassing a large database of gait and speeds. This replete database offers the solid background to train and test the machine learning classifiers for the assessment of gait.

## Research Strategy

The first one of the outlined research steps is known as systematic pre-processing, analysis, and subsequent classification of gait data by applying multiple machine learning techniques. The research will be conducted in several phases:The research will be conducted in several phases:

1. Data Preprocessing:

• Preprocess the data by loading the data and then cleaning the data for any unwanted features or any noise.

• This helps to put the data in a standard format and for the purpose of engaging in precise analysis.

• Randomly divide the dataset to get the training as well as the testing sets to compare the performance of classifier.

2. Feature Extraction and Selection:

• Identify parameters that can be obtained from the kinematic data, position and orientation, stride length, temporal parameters etc.

• Select the most important features that help one distinguish between the different gait types using a method like PCA and using feature importance analysis.

3. Classifier Implementation:

• constantly which machine learning classifiers are Support Vector Machine (SVM), Naive Bayes, Decision Tree, Random Forest, Logistic Regression and K-Nearest Neighbors (KNN).

• Split each set by using cross validation on the training dataset and test the classifiers on the testing dataset.

4. Performance Evaluation:

• Measures that are commonly used in classification problems including accuracy, precision, recall and F1-score should be applied in order to assess the performance of each classifier.

• It has to do cross validation to check and ascertain that the classifiers are accurate and reliable.

5. Comparison and Analysis:

• Then, compare the results of the different classifiers to determine which classification model is the best performimg in distinguishing gait patterns.

• Using the results determine the advantages and disadvantages of each classifier as well as patterns in gait data.

# Tools and Techniques

The project will utilize various tools and techniques to preprocess, analyze, and classify the gait data:The project will utilize various tools and techniques to preprocess, analyze, and classify the gait data:

1. Programming Languages:

• Python will be the primary programming language because of the wide availability of libraries and frameworks available for data analysis and machine learning.

2. Libraries and Frameworks:

• Pandas and NumPy: This is mostly used in computations, in the sense of data manipulation operations performed over numbers.

• Scikit-learn: It will also be useful for applying and assessing machine learning classifiers.

• Matplotlib and Seaborn: Under data preparation and visualization in order to gather more information about the data and its nature.

3. Machine Learning Algorithms:

• Support Vector Machine (SVM): Applicable in cases of high dimensions, and is not sensitive to the phenomenon of overlearning.

• Naive Bayes: A naive but very effective probabilistic classifier that uses Bayes' theorem as the basis of its prediction.

• Decision Tree: It is a non-parametric model that is easy to understand and usually accompanied by excellent visuals.

• Random Forest: The type of ensemble learning wherein the different decision trees are used successively to reduce the error rate of the decision trees and avoid overfitting.

• Logistic Regression: A binary classifier that works suitable for binary classification issues.

• K-Nearest Neighbors (KNN): A type of learning algorithm that operates through the utilization of nearest neighbors and reputation while categorizing based on the most frequently occurring class of the nearest neighbors.

Thus, the goal of this project is to work on the classified features of gait analysis that will advance the filed of biomechanics and the use of machine learning.

# Expected Outcomes

For the current project, it is expected that the identification of distinct gait patterns and/or speeds is feasible by building up effective machine learning classifiers. The classifiers that will be assessed are accuracy, precision, recall, and F1-score. Thus, the project's goal is to compare different classifiers, where the most suitable methods for gait analysis will be determined. The expected result will be developed and proved classifiers which can be used in clinic diagnosis, rehabilitation and in sports science field. Moreover, the project will be important for evaluating the effectiveness of various methods that belong to the machine learning field for gait analysis.

# Conclusion

This project sees the need to contribute to the enhancement of gait analysis by improving the development of classifiers through the use of machine learning. The proposed methodology of data preprocessing, feature choice and extraction, implementation of the classifiers, and evaluation of the performances will provide a detailed analysis and accurate results. Consequently, the successful achievement of the proposed work will contribute to the development of new diagnostic and therapeutic instruments useful in clinical applications and to the optimisation of performance evaluation in sports science.

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